

Shashi B. Mishra

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EDUCATION

Ph.D. in Physics

March 2021

Indian Institute of Technology Madras, India

Thesis: *Ab initio* Modelling and Functionalization of Graphene & TiO₂ Surfaces and Interfaces

Advisor: Prof. B. R. K. Nanda

M.Sc. in Physics

June 2013

Utkal University, Odisha, India

B.Sc. in Physics

June 2011

Fakir Mohan University, Odisha, India

PROFESSIONAL APPOINTMENTS

Assistant Research Scientist

March 2026 -continuing

Department of Electrical and Computer Engineering, University of Maryland, MD, USA

Working with: Prof. Thomas E. Murphy & Prof. Pratibha Dev

Postdoctoral Fellow

April 2023 - March 2026

Department of Physics, SUNY Binghamton, NY, USA

Advisor: Prof. Roxana Margine

Postdoctoral Fellow

April 2021 - March 2023

Department of Mechanical Engineering, University of California, Riverside, USA

Advisor: Prof. Sinisa Coh

TEACHING EXPERIENCE

Guest Lecturer, Statistical Mechanics I (PHYS 631), Graduate courses, Physics Dept., Binghamton University, Fall 2025

Teaching Assistant, Department of Physics, IIT Madras

2015-2019

- PH1030 Physics Lab, Undergraduate Phys- Jul-Nov 2015
- PH1020 Physics II, Undergraduate Phys- Jan-May 2016
- EP3291 Engineering Physics Lab, Jul-Nov 2016
- PH5060 Physics Lab, Master in Physics, Jan-May 2017
- PH1010 Physics I, Undergraduate Phys, Jul-Nov 2017
- PH5770 Advanced Physics Lab, MSc, Jan-May 2018

TECHNICAL SKILLS

Software: Quantum ESPRESSO, VASP, EPW, Wannier90, WannierBerri, WannierTools, Gaussian

Programming: FORTRAN, Python, MATLAB, Bash

HPC: Experience with parallel computing, job scheduling (SLURM), optimization

RESEARCH INTERESTS

- DFT / MBPT
- Defects and Quantum computing
- Electron-phonon interactions
- Light-matter interaction
- Superconductivity
- Machine learning
- Strongly correlated electrons
- Topological quantum materials
- Magnetism & spintronics
- Transport theory
- Energy materials
- Catalysis

RESEARCH FUNDING

- **Principal investigator**, PHY250052 (2025-2026): ACCESS computational resources, 150,000 credit hours

HONORS AND AWARDS

- APS DCOMP Travel Award 2026 to attend the Global Physics Summit 2026 in Denver, Colorado
- Young Scientist Award 2024, [Odisha Physical Society](#), Odisha, India
- **Best Ph.D. Thesis Award in Physics 2021** (Prof A L Lashkar Prize), IIT Madras, India
- DST Young Scientist Travel Award (2019) to attend the international conference at ICAMM, Rennes, France
- Oral presentation award, EESTER International Conference, SRM and IIT Madras (2020), India
- Best poster, International Conference on Advanced Functional Materials (ICAFM), Anna University (2017), India

- First Prize, I2CAM School on Clean and Renewable Energy, JNCASR, Bangalore (2017), India
- Meritorious Master Scholarship, Institute of Mathematics & Applications Scholarship, DST India (2011-2013)

PUBLICATIONS

1. **S. B. Mishra***, Z. Liu*, Sabyasachi Tiwari, Feliciano Giustino, and E. R. Margine, "Comparative study of phonon-limited carrier transport in the Weyl semimetal TaAs family" *Phys. Rev. B* **113**, 045202 (2026).
2. **S. B. Mishra**, and R. Margine, "Nonadiabatic and anharmonic effects in high-pressure H₃S and D₃S superconductors" *Annalen der Physik* **538**, e00553 (2026).
3. S. Dey*, **S. B. Mishra***, A. Hazra, D. Kabiraj, and S. C. Roy, "Anomalous gas sensing characteristics of TiO₂ nanorod arrays irradiated with high-energy ion beam" *Mater. Sci. Eng. B*, **326**, 119210 (2026).
4. **S. B. Mishra**, H. Mori, and E. R. Margine, "Electron-phonon vertex correction effect in superconducting H₃S" *npj Comp. Mat.*, **11**, 342 (2025).
5. **S. B. Mishra**, "Inverse Faraday effect in 3d, 4d, and 5d transition metals," *Phys. Rev. B* **111**, 174413 (2025).
6. Z. Liu*, **S. B. Mishra***, J.-M. Lihm, S. Poncé, and E. R. Margine, "Phonon-limited carrier transport in the Weyl semimetal TaAs," (*equal contribution) *Phys. Rev. B* **112**, 104311 (2025).
7. Y. Rambabu, **S. B. Mishra**, O. Henrotte, and E. Cortes, "Defects Dynamics in Photo-Excited CeO₂ and their Influence on CO₂ Photoreduction", *Adv. Funct. Mater.*, e13933 (2025).
8. X. Zhang, **S. B. Mishra**, E. R. Margine, and E. Kioupakis, "Cubic BeB₂: A metastable *p*-type conductive material from first principles", *Phys. Rev. B* **112**, 155206 (2025).
9. A. Das, S. Gumber, N. C. Maji, **S. B. Mishra**, Preethi M., P. Ning, R. G. Nair, Abhijith T, A. S. Vasenko, Madhumitha R., and O. V. Prezhdo, "Electrostatic Self-assembly Driven Heterojunction of CeO₂/g-C₃N₄ Nanosheets for Efficient Photocatalytic Hydrogen Evolution", *ACS Applied Materials & Interfaces* (2025).
10. **S. B. Mishra**, E. T. Marcial, S. Debata, A. N. Kolmogorov, and E. R. Margine, "Stability-superconductivity map for compressed Na-intercalated graphite," *Phys. Rev. B* **110**, 174508 (2024).
11. P. Sahoo, **S. B. Mishra**, S. Debata, A. Sharma, B. Sahu, S. Padhan, R. Thangavel, B-K. Lee, "Hydrothermally Grown Halogen-Doped ZnO Nanorods for Photoelectrochemical Water Oxidation: Experimental and DFT Insights," *J. Phys. Chem. C* **128**, 18711 (2024).
12. Abhijitha V. G., **S. B. Mishra**, and B. R. K. Nanda, "Si₂BN nanosheets as anchoring cathode material for realizing high capacity Al-ion battery," *J. Energy Storage* **77**, 109913 (2024).
13. V. Ortiz, **S. B. Mishra**, L. Vuong, S. Coh, and R. B. Wilson, "Transient Ellipsometry Measurements of the Specular Inverse Faraday Effect in Transition Metals," *Phys. Rev. Mater.* **7**, 125202 (2023).
14. **S. B. Mishra** and S. Coh, "Spin contribution to the inverse Faraday effect of nonmagnetic metals," *Phys. Rev. B* **107**, 214432 (2023).
15. B. Biswal, **S. B. Mishra**, R. Yadav, S. Poudyal, R. Rajarapu, P. K. Barman, K. R. Pandurang, M. Mandal, R. Singh, B. R. K. Nanda, and A. Misra, "Work Function of van der Waals Topological Semimetals," *Appl. Phys. Lett.* **120**, 093101 (2022).
16. Abhijitha V. G., **S. B. Mishra**, S. Ramaprabhu, and B. R. K. Nanda, "Design of Aluminium ion battery with Graphyne host," *Nanoscale Adv.* **4**, 3870-3882 (2022).
17. S. Dey, A. Chakravorty, **S. B. Mishra**, N. Khatun, A. Hazra, B. R. K. Nanda, S. Chandran, D. Kabiraj, and S. C. Roy, "High energy ion induced modification in TiO₂ nanorods" *Nanoscale Adv.* **4**, 241-249 (2022).
18. **S. B. Mishra**, Abhijitha V. G., S. Ramaprabhu, and B. R. K. Nanda, "Graphdiyne-A promising Cathode Material for Aluminium dual ion Battery," *ACS Appl. Energy Mater.* **4**, 7786-7799 (2021).
19. **S. B. Mishra**, S. Marutheeswaran, S. C. Roy, V. Natarajan, P.K. Rai, and B. R. K. Nanda, "Adsorption and Degradation Mechanism of 2,4,6-TNT on TiO₂ (110) surface," *Surf. Sci.* **713**, 171902 (2021).
20. J. Khatua, T. Arh, **S. B. Mishra**, H. Luetkens, A. Zorko, B. Sana, M.S.R. Rao, B. R. K. Nanda, and P. Khuntia, "Development of Short and Long-range Magnetic order in Double Perovskite Ba₂MnTeO₆," *Sci. Rep.* **11**, 6959 (2021).
21. **S. B. Mishra**, S. C. Roy, and B. R. K. Nanda, "Electronic Structure of Graphene/TiO₂ Interface: Design and Functional Perspectives," *Appl. Surf. Sci.* **542**, 148709 (2021).

22. **S. B. Mishra**, S. K. Yadav, D. G. Kanhere, and B. R. K. Nanda, "Fluorine Intercalated Graphene: Formation of a Two-dimensional Spin Lattice through Pseudoatomization," *Phys. Rev. Mater.* **4**, 074411 (2020).
23. **S. B. Mishra** and B. R. K. Nanda, "Facet Dependent Catalytic Activities of Anatase TiO₂ for CO₂ Adsorption and Conversion," *Appl. Surf. Sci.* **531**, 147330 (2020).
24. P. Panigrahi*, **S. B. Mishra***, T. Hussain, B. R. K. Nanda, and R. Ahuja, "Density Functional Theory Studies of Si₂BN Nanosheets as Anode Materials for Magnesium Ion Batteries," *ACS Appl. Nano Mater.* **3**, 9055 (2020).
25. S. Marutheeswaran*, **S. B. Mishra***, S. C. Roy, and B. R. K. Nanda, "Mechanistic Understanding of NO₂ Dissociation on a Rutile TiO₂(110) Surface: An Electronic Structure Study," *J. Phys. Chem. C* **124**, 8786 (2020).
26. **S. B. Mishra**, A. Choudhary, S. C. Roy, and B. R. K. Nanda, "Quantum-Mechanical Process of Carbonate Complex Formation and Large-scale Anisotropy in the Adsorption Energy of CO₂ on Anatase TiO₂(001) Surface," *Phys. Rev. Mater.* **2**, 115801 (2018).
27. S. Samanta, **S. B. Mishra**, and B. R. K. Nanda, "Quantum well Structure of Double Perovskite Superlattice and Formation of Spin-polarized Two-dimensional Electron gas," *Phys. Rev. B* **98**, 115155 (2018).

Manuscripts under review:

1. M. Keivanloo, F. D., **S. B. Mishra**, M. S, M. J. A, M. Kawamura, E.R. Margine, H. R., K. H., R. M., and M. Khazaei, "Superconductivity of surface functionalized two-dimensional o-MXenes" *arXiv:2602* (2026) [NPJ Comput. Mater.].
2. Zahir Muhammad*, **S. B. Mishra***, Gayatri, Grzegorz Krasucki, Wajid Ali, Obaid Iqbal, Zia Ur Rehman, Aziz Ur Rahman, Maciej R. Molas, Lin Xiaoyang, Weisheng Zhao, "Chiral Phonons and Giant Photoresponse in anisotropic Quasi-1D van der Waals Semiconductor ZrSnS₃" [Adv. Funct. Mater.]

Manuscripts in preparation:

1. S. Tiwari, J. Lafuente-Bartolome, **S. B. Mishra**, M. Zacharias, S. Poncé, E. Kioupakis, E.R. Margine, and F. Giustino, "Automation and Abstraction for Many-Body Physics: Bridging Complexity using EPWpy".
2. **S. B. Mishra**, R. Margine, and R. Akashi, "Role of higher-order van Hove singularity effect on self-energy of H₃S"
3. **S. B. Mishra**, Francesco Belli, Eva Zurek, and R. Margine, "Effect of non-adiabatic electron-phonon interaction and anharmonicity on superconducting properties of high-pressure hydrides."

Conference proceedings: **S. B. Mishra** and B. R. K. Nanda, "Virtual Synthesis of Crystals using *Ab initio* MD: Case study on LiFePO₄," *AIP Conf. Proc.* **1832**, 090044 (2017).

PROFESSIONAL SERVICE

- **Treasurer/Secretary**, APS Topical Group on Energy Research and Applications (GERA) (2025-2027)
- **NSF CSSI PI Meeting**: Represented at NSF CSSI-PI meeting in Denver, CO (July 2025)
- Early career representative, APS Division of Computational Physics (DCOMP) (2024-2025)
- **APS Annual Leadership Meeting and Congressional Visit**: Representative for DCOMP, DC (January, 2025)
- **Session Chair**, APS March Meeting 2024: "Session K58: Electrons, Phonons, and Electron-Phonon Scattering"
- **Peer Reviewer**: *Phys. Rev. Lett.*, *Phys. Rev. Applied*, *Phys. Chem. Chem. Phys.*, *Sci. Rep.*, *J. Energy Storage*, *Int. J. Hydrogen Energy*, *Comp. Phys. Comm.*, *Chem. Eng. Sci.*, *Phys. Lett. A*, *Solid State Comm.*, *Comput. Theor. Chem.*, and *J. Phys. Chem. Solids*.

SELECTED PRESENTATIONS

Invited Talks

- "Insights into the surface chemistry of oxide semiconductors," *ICAM-SEEd 2025 Workshop*, IITRAM, Ahmedabad, India (November 27, 2025)
- "Understanding phonon-mediated superconductivity: From fundamentals to frontiers," *International Faculty Development Program on Frontiers in Material Science*, SRM IST, Chennai (March 21, 2025)
- "Phonon-mediated superconductivity in H₃S and NaC₄," *Young Scientist Award Winner Talk, 41st Annual Convention of Orissa Physical Society*, KIIT University, Bhubaneswar (Invited Award Talk, February 8, 2025)
- "An Introduction to Electronic Properties of Materials and Magnetism in Graphene," *Department of Physics Seminar, Fakir Mohan Autonomous College*, Balasore, India (January 25, 2019)

Contributed Presentations (2 of 28 total)

- "Automating Many-Body Workflows with EPWpy in the MATCSSI Framework," *NSF CSSI-PI Meeting*, Denver (2025)
- "Stability-superconductivity map for compressed Na-intercalated graphite," *APS March Meeting*, Anaheim (2025)